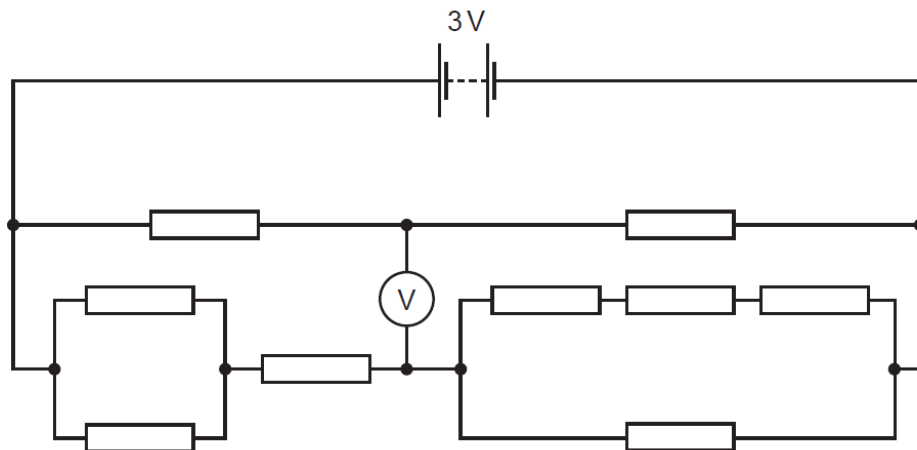


22 A circuit is set up as shown, supplied by a 3 V battery. All resistances are 1 k Ω .



What will be the reading on the voltmeter?

A 0 V

B 0.5 V

C 1.0 V

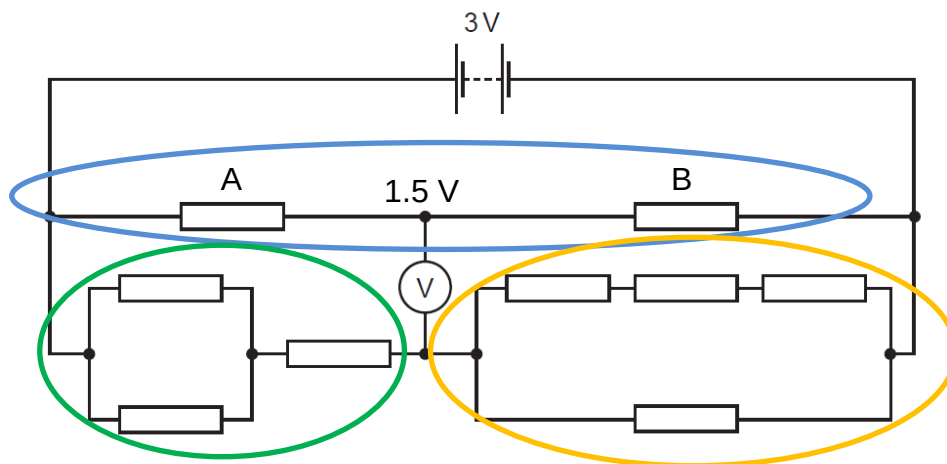
D 1.5 V

Answer: B

Resistors A and B (in blue circle) have a p.d. of 3 V across them.

Since resistors A and B have the same resistance $\rightarrow$ 1.5 V across each of them.

Top end of voltmeter will have a potential of 1.5 V since $3 - 1.5 \text{ V} = 1.5 \text{ V}$.



Effective resistance of resistors in green circle = $\frac{1}{2} + 1 = 1.5 \text{ k}\Omega$.

Effective resistance of resistors in yellow circle $= \left(\frac{1}{1+1+1} + 1 \right)^{-1} = 0.75 \text{ k}\Omega$.

There is a p.d. of 3 V across resistors in green and yellow circles.

p.d. across resistors in green circle $= \left(\frac{1.5}{1.5 + 0.75} \right) (3) = 2 \text{ V}$.

Potential at lower end of voltmeter $= 3 - 2 = 1 \text{ V}$.

Reading on voltmeter will be the p.d. across top and bottom ends $= 1.5 - 1 = 0.5 \text{ V}$.